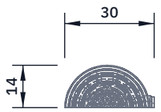
CONCEPT, NOT FOR FABRICATION

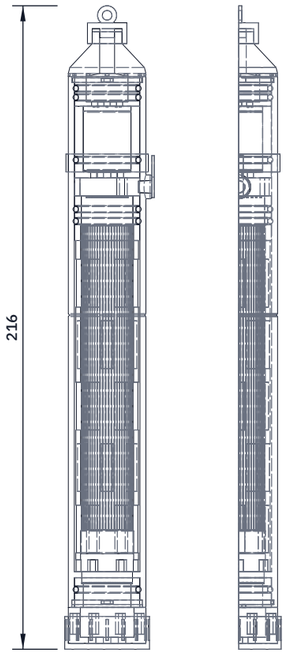
Helpful Water A: filter straw · github.com/BoujeeEnjinia1701/helpful-water

REV	DESCRIPTION	DATE	BY
P1	Preliminary half section from the TRL 3 model	2026-11-11	AC
P2	Air check restated after the independent STEM review	2026-11-11	AC



TOP VIEW

Scale 1:2; looking down (along -Z)



FRONT VIEW

Scale 1:2; looking at the front (along +Y)

RIGHT VIEW

Scale 1:2; looking at the right side (along -X)



ISOMETRIC VIEW

Half section, front half removed; not to scale; seen from the front right and above, about 30 deg elevation

FLOW PATH AND CLEAN/DIRTY BOUNDARIES

Dirty side: intake cap slots, mesh, bay, around the fibres
Boundary 1: fibre walls (0.1 or 0.2 um rating, to decide)
Boundary 2: top potting, bought pre-potted module
Boundary 3: two O-rings on the module collar
Clean side: lumens, header 150 to 156, carbon, mouthpiece
Mouthpiece: two O-rings; joint kept above the dip line
Backflush: plug out, cap on; syringe into the socket;
water runs header, lumens, fibre walls, out of the intake
Carbon dead-ended by the cap plug during backflush
Air check: syringe to the 6 mL mark (50 kPa), plunger holds
PRELIMINARY, NOT FOR FABRICATION

HELPFUL WATER A: FILTER STRAW

github.com/BoujeeEnjinia1701/helpful-water

Hollow-fibre filter straw (option A1): half section through the flow path and seals

DRAWING NO.	REV	SHEET	SIZE
HWA-DWG-002	P2	1 of 1	ANSI B

SCALE	UNITS	PROJECTION	DATE	DRAWN
1:2	mm		2026-11-11	Amish Chadha

MATERIAL / NOTES

PP parts (printed for the prototype); 316L mesh; EPDM O-rings; bought fibre module; GAC. See bom/bom.csv

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